L1 Blockchain Development Plan (Stage 1)

Goal: Build a Substrate-based L1 blockchain that supports validators, staking, governance, gas model, smart contracts (Ink! + EVM), and robust infrastructure for future stablecoin deployment.

Phase 1 – Core Architecture & Setup

Tasks:

- Project setup (Rust + Substrate environment)
- Define chain spec (network name, token, decimals, genesis config)
- Implement consensus (BABE + GRANDPA)
- Add NPoS staking pallet
- Setup transaction fees (gas model)
- Initial testing on local single-node devnet

Deliverables:

- Running local node
- Documentation for setup and build
- GitHub repo initialized

Testing:

- Block production
- Validator registration and staking logic
- Transaction fee flow

Phase 2 – Validator Network & Governance

Tasks:

- Multi-validator setup on testnet
- Enable on-chain governance (democracy, council, technical committee pallets)
- Configure inflation and staking reward logic
- Integrate validator onboarding and slashing parameters

Deliverables:

- Public testnet with 3–5 validators
- Governance proposals functional
- Staking and rewards active

Testing:

- Validator join/leave
- Governance proposals and voting
- Inflation & slashing logic

Phase 3 – Smart Contract Layer

Tasks:

- Add smart contract support (Ink! and EVM-compatible runtime)
- Implement multi-token support (mint/burn/oracle-ready logic)
- Test with sample token smart contracts
- Performance tuning for 3,000–10,000 TPS target

Deliverables:

- Smart contract execution functional
- Token mint/burn working on testnet
- Benchmarks for TPS and gas cost

Testing:

- Contract deployment and execution
- Multi-token transactions
- TPS & gas measurement

Phase 4 – Infrastructure & Monitoring

Tasks:

- Deploy RPC nodes (public & private)
- Implement monitoring with Prometheus + Grafana
- Setup daily snapshots & backup system
- Integrate explorer (React + Subsquid backend)
- Add wallet connection (Polkadot.js + Web wallet API)

Deliverables:

- Running testnet infrastructure
- Explorer & wallet connected
- Live metrics dashboard

Testing:

- RPC response
- Explorer indexing
- Wallet transaction flow

Phase 5 – QA, Documentation & Launch Prep

Tasks:

- Full integration testing
- Runtime and staking unit tests

- Governance upgrade simulation
- Write detailed documentation

Deliverables:

- Audit-ready code
- Full documentation: architecture, validator onboarding, RPC, explorer, wallet APIs
- Ready for mainnet deployment

Testing:

- Regression testing
- On-chain upgrade test
- Validator rotation stability